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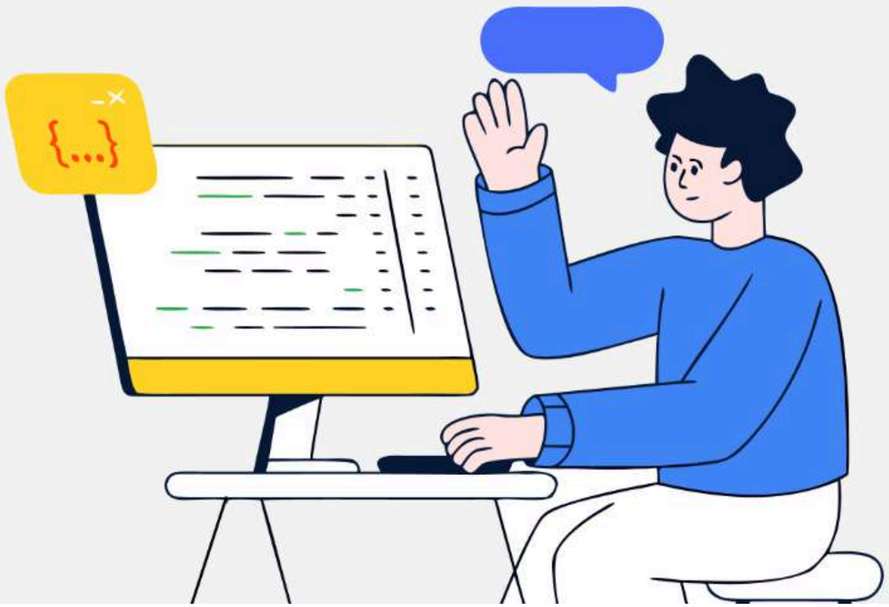
Solution Challenge



Team Details

a. **Team name:** RayCtify
b. **Team leader name:** Ishana Datta
c. **Problem Statement:** Unbiased AI Decision - Financial institutions lack a secure, automated way to detect and neutralize systemic AI bias without exposing sensitive data or undertaking costly model retraining

In India, getting a loan can shape someone's entire future - whether it's starting a business, paying for education, or supporting a family.
But today, many of these decisions are made by AI systems.

And as the Reserve Bank of India has warned, 'AI systems trained on biased data can perpetuate discrimination.'

What makes it worse is - many applicants don't even know why they were rejected.

That means bias isn't always visible - but its impact is very real.

And that's exactly the problem RayCtify is built to solve.

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SOLUTION BRIEF

RayCtify: Enterprise AI Bias Remediation

The Problem: Financial ML models inherit systemic bias, forcing institutions to choose between discriminatory lending or discarding multi-million-dollar production models.

Our Solution: RayCtify is an end-to-end algorithmic compliance framework that doesn't just detect bias- it mathematically heals it while strictly preserving financial accuracy.

The Auditor & Arena (Detection): Uses Counterfactual Testing to prove bias. It flips an applicant's demographic data while keeping financial merit identical, visually exposing hidden penalties through real-time, side-by-side model benchmarking.

The Interceptor (Instant Triage): A post-processing engine that analyzes biased production models, calculates their exact demographic penalty, and bolts a mathematical fairness filter onto the output layer to instantly heal decisions on the fly.

The Vaccine (Root-Cause Cure): A data inoculation engine that parses historical datasets, calculates the Disparate Impact Ratio (DIR), and injects the exact minimum dose of mathematically perfect "Counterfactual Twins" to permanently neutralize bias in future training cycles.

The Impact: We transform AI ethics into an automated engineering pipeline, ensuring that loan approvals are driven entirely by mathematical meritocracy, not demographic proxies.

OPPORTUNITIES

How different is it from any of the other existing ideas?

How will it be able to solve the problem?

USP of the proposed solution

HOW DIFFERENT IS IT FROM ANY OF THE OTHER EXISTING IDEAS?

Unlike existing open-source libraries (which require complex coding) or enterprise cloud solutions (which require uploading sensitive data), RayCtify is a zero-retention, client-side application. While other tools like IBM AIF360 or AWS SageMaker Clarify exist, they are built for data scientists sitting in a lab. **RayCtify is built for the high-stakes, high-security world of enterprise finance.**

Traditional Retraining

01 **Data Security Risk**: Requires uploading highly sensitive, proprietary financial datasets to external cloud servers for processing.

02 **High Cost & Time**: Demands completely scrapping and retraining the model from scratch, costing significant time and compute resources.

03 **Code-Heavy Barrier**: Relies on complex open-source libraries, restricting use strictly to specialized data scientists and engineers.

04 **Manual Remediation**: Forces teams into guesswork - dropping columns or tweaking data manually with no guarantee of parity.

05 **No Live Benchmarking**: Lacks real-time visual tools to compare the old biased model against the newly adjusted one.

The RayCtify Approach

01 **100% Zero-Retention**: Executes all file processing and PDF/Model generation entirely client-side. Data never leaves the local machine.

02 **Instant Remediation**: Uses a mathematical "Fairness Wrapper" to fix discriminatory decisions on the fly - zero retraining required.

03 **No-Code Accessibility**: Provides an intuitive visual dashboard, allowing business stakeholders and auditors to test models instantly.

04 **Automated Precision**: Automatically calculates the Disparate Impact Ratio and synthesizes the exact mathematical dose of fair data needed.

05 **Live Visual Benchmarking**: Uses "The Arena" to run synchronized applicant data, instantly highlighting the exact "Bias Delta" between models.

HOW WILL IT BE ABLE TO SOLVE THE PROBLEM?

1. **The Check-up (The Auditor)**

Most AI is a "black box" - you don't know why it says "No." We use The Auditor to run "What If" scenarios locally in your browser.

- The Standout: We use Gemini 3 Flash to explain the bias in plain English. Instead of just showing a scary graph, it tells the bank: "Your model is 20% more likely to reject this person simply because of their zip code."
- The Unique Part: This happens entirely in your browser. We don't steal your data; we audit it locally.

2. The Instant Fix (The Interceptor)

This is our biggest "Flex." Instead of retraining the model (which takes months), we use The Interceptor. We bolt a "Fairness Wrapper" onto the existing model.

- The Standout: It’s an Instant Cure. It catches a biased decision as it's happening and mathematically adjusts it to be fair based on merit.
- The Unique Part: No retraining required. You keep your expensive model, but you make it ethical instantly.

3. The Side-by-Side Proof (The Arena)

To trust the fix, you need to see it. The Arena lets you run the same group of people through the Biased model and our Healed model at the same time.

- The Standout: We use Google Stitch to make this a beautiful, high-fidelity dashboard. You can visually see the "Bias Delta" (the gap) disappear in real-time.
- The Unique Part: It proves to regulators and lawyers that the AI is now fair without losing any financial accuracy.

4. Permanent Immunity (The Vaccine)

To make sure the next model isn't born with the same bias, we use The Vaccine.

- The Standout: It uses Gemini 3 Pro to study the historical data and synthesize "Counterfactual Twins." These are mathematically perfect data points that "teach" the AI what a fair world looks like.
- The Unique Part: We don't just fix the symptoms; we vaccinate the data so the next generation of AI is born "immune" to bias.

USP OF THE PROPOSED SOLUTION

1. Zero-Retention Architecture (The Security Standout)

- The Problem elsewhere: Existing cloud solutions (AWS/Azure) require banks to upload sensitive, proprietary financial data to external servers. This is a massive legal and security risk.
- The RayCtify Fix: Our architecture is Stateless. Using FastAPI and PapaParse, we shred and process data entirely on the client-side. The data never lives in a database; it never leaves the bank's machine. We provide "Zero-Trust" auditing that legacy tools can't touch.

2. The "Interceptor" Wrapper (The Cost Standout)

- The Problem elsewhere: When a model is biased, most tools say, "Scrap it and retrain." This costs millions in compute power and weeks of engineering time.
- The RayCtify Fix: We don't retrain. We "intercept." Our Interceptor module bolts a mathematical Fairness Wrapper (Equalized Odds) onto the existing model. It heals the decisions as they happen without needing to change a single line of the original model's code.

3. Counterfactual Inoculation (The Future-Proof Standout)

- The Problem elsewhere: Existing solutions just "debias" by dropping columns (like race or gender), which actually makes the model less accurate because of proxy variables.
- The RayCtify Fix: The Vaccine. Our Star feature. Instead of deleting data, we use Gemini 3 Pro to synthesize "Counterfactual Twins." We add mathematically perfect, fair data points to the training set. This doesn't just fix the current model; it "vaccinates" future models so they are born immune to bias.

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LIST OF FEATURES

01 Zero-Retention Security Architecture

Engineered as a stateless system where all sensitive file parsing (CSVs, Models) and report generation happen entirely on the client-side. Leverages PapaParse and the Browser Blob API to ensure proprietary financial data never touches a database or external cloud storage.

02 The Auditor (Counterfactual Testing)

Allows users to input specific demographic and financial parameters to test for discriminatory outcomes in real-time. Generates a secure, client-side PDF Audit Report that documents the model's decision-making patterns for regulatory review.

03 The Arena (Comparative Benchmarking)

Simultaneously runs inference on both the "Reference" (biased) and "Test" (healed) models using identical datasets. A high-fidelity dashboard that highlights the exact mathematical gap in approval rates between different demographic groups.

04 The Interceptor (Instant Remediation)

Applies Equalized Odds Post-Processing math to existing models, neutralizing bias at the output layer. Allows users to download a "RayCtified" .pkl file - a healed version of their production model that is ready for deployment instantly.

05 The Vaccine (Data Inoculation)

Automatically calculates the Disparate Impact Ratio of historical datasets to identify root-cause bias. Generates a mathematically optimized "inoculation dose" of fair data points and appends them to the original dataset. Provides a downloadable, bias-neutral .csv file to ensure future training cycles are meritocratic.

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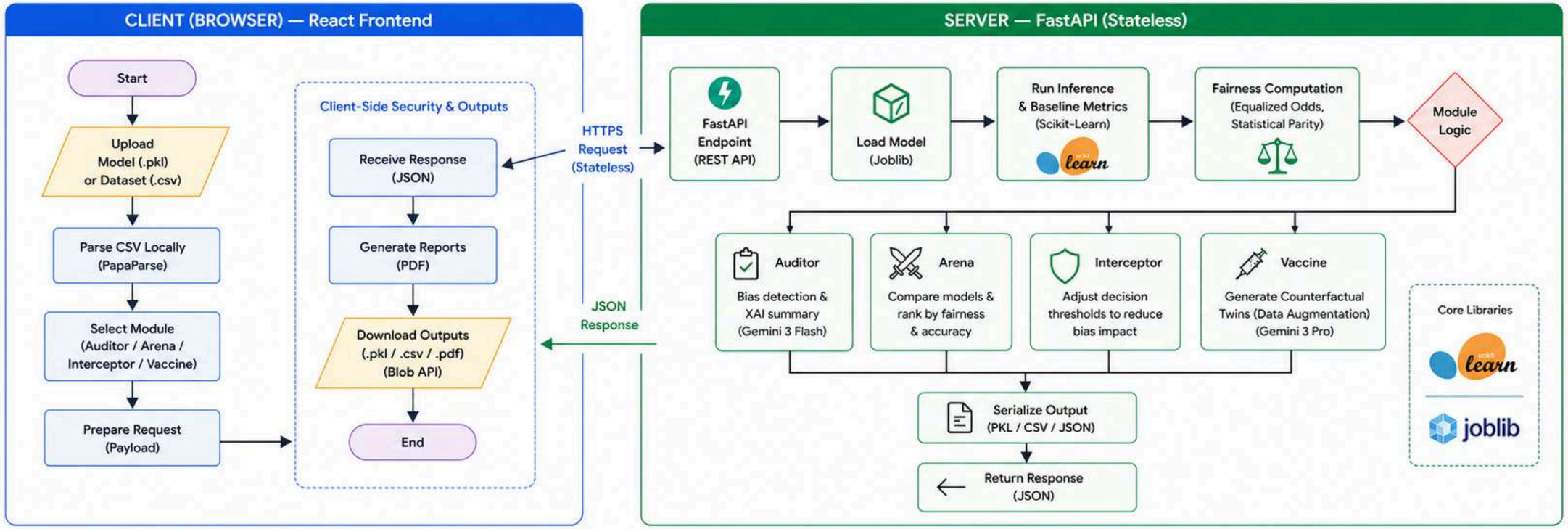
PROCESS FLOW DIAGRAM



Intelligence & Generative AI

- Gemini 3 Pro (Vaccine – Counterfactual Generation)
- Gemini 3 Flash (Auditor – XAI Summaries)
- Google AI Studio (Prompt Engineering & Tuning)

Development & Design Ecosystem

- Google Antigravity (Agentic IDE) (Refactor & Unit Tests)
- Google Stitch (Labs) (Vibe Design – UI with Voice Canvas)

Frontend Technologies (Client)

- React (TypeScript) (UI Framework)
- Tailwind CSS (UI Styling)
- PapaParse (Local CSV Shredding)
- Browser Blob API (Secure Downloads)

Backend Technologies (Server)

- FastAPI (Python) (Web Framework)
- Scikit-Learn (ML & Fairness Math)
- Joblib (Model Load / Dump)

Infrastructure & Deployment (Google Stack)

- Google Cloud Run (Backend Hosting)
- Firebase Hosting (Frontend Hosting)
- Google Cloud Logging (Monitoring – No Payload)

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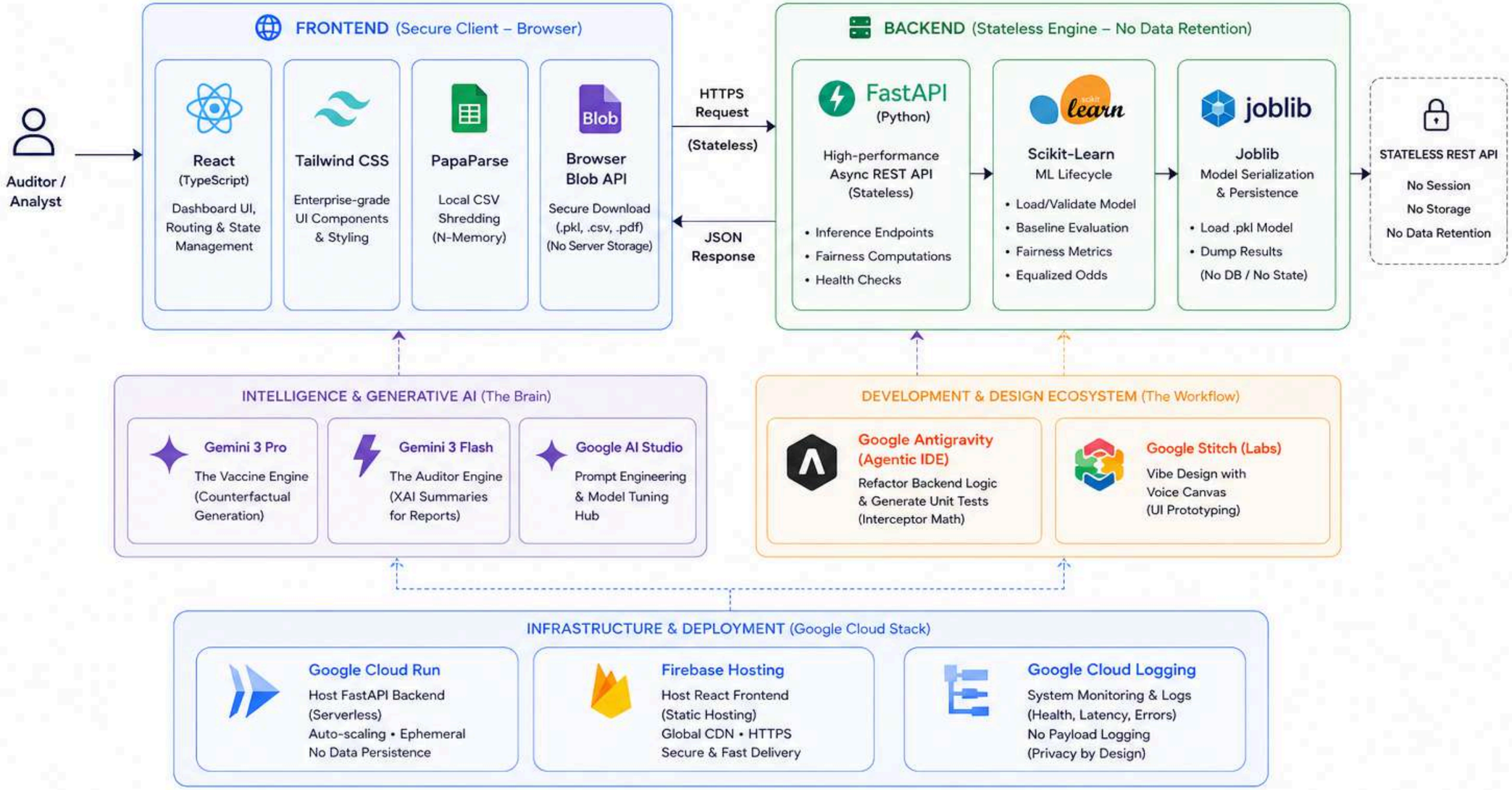
ARCHITECTURE DIAGRAM OF THE PROPOSED SOLUTION



TECHNOLOGIES TO BE USED IN THE SOLUTION

1. Intelligence & Generative AI (The Brain)

- Gemini 3 Pro: The primary reasoning engine used for The Vaccine. It handles the deep mathematical synthesis required to generate "Counterfactual Twins" that are statistically perfect.
- Gemini 3 Flash: Powering The Auditor. It provides lightning-fast, real-time "Explainable AI" (XAI) summaries for the PDF audit reports, translating complex bias data into human-readable text.
- Google AI Studio: Our centralized hub for prompt engineering and model tuning, used to refine the system instructions for the Gemini API.

2. Development & Design Ecosystem (The Workflow)

- Google Antigravity (Agentic IDE): Our primary development platform. We utilized Antigravity Agents to autonomously refactor the "Interceptor" fairness math and generate the comprehensive unit test suite for the FastAPI backend.
- Google Stitch (Labs): Used for "Vibe Design." We leveraged Stitch's Voice Canvas to prototype the entire UI. By describing the "Arena" dashboard verbally, we generated the production-ready React + Tailwind CSS frontend code instantly.

3. Frontend Architecture (The Secure Client)

- React (TypeScript): The foundation of the dashboard, ensuring a type-safe and high-performance user experience.
- Tailwind CSS: For the enterprise-grade, "Build with AI" branded visual interface.
- PapaParse: The engine for local CSV shredding, ensuring all sensitive financial data is parsed strictly in the browser memory.
- Browser Blob API: Manages the secure, client-side creation and download of the "RayCtified" .pkl models and .csv datasets.

4. Backend Architecture (The Stateless Engine)

- FastAPI (Python): A high-performance, asynchronous web framework that handles all ML inference and fairness calculations.
- Scikit-Learn / Joblib: Used for the core ML lifecycle - loading biased models, running baseline tests, and executing the Equalized Odds math.
- Stateless REST API: Designed with no internal state or database to maintain the Zero-Retention security mandate.

5. Infrastructure & Deployment (The Google Stack)

- Google Cloud Run: Hosts the FastAPI backend. It provides serverless scaling and ensures that no user data persists in the environment after a request is completed.
- Firebase Hosting: Deploys the React frontend globally, providing a secure, SSL-encrypted gateway for auditors.
- Google Cloud Logging: Used strictly for monitoring system health and latency without capturing any payload data.

ESTIMATED IMPLEMENTATION COST

1. **Current Development Scale (Low-to-Zero Cost)**

Based on our current tech stack (GCP, Firebase, React), our monthly expenses are ₹0 (completely within free tiers) if we stay under specific usage limits.

- Hosting: Firebase Hosting (Spark Plan) covers our frontend for free up to 10GB of storage.
- Backend: Google Cloud Run (us-central1) gives us the first 180,000 vCPU-seconds and 360,000 GiB-seconds free every month.
- Builds: Google Cloud Build offers 2,500 build-minutes free per month, more than enough for your current CI/CD pipeline.

2. Startup/Small Business Scale (~₹1,500 - ₹5,000 / month)

If we were to launch this as a specialized "Audit-as-a-Service" for boutique fin-techs, our operational costs would shift to the Google Cloud Blaze (Pay-as-you-go) plan:

- Compute: Cloud Run for an active auditing service (running ~10% of the month) would cost roughly $10–$15 (₹800–₹1,250).
- Storage: Artifact Registry costs $0.10 per GB for storing our Docker images beyond the first 0.5 GB.
- Bandwidth: Standard egress (data leaving GCP to our users) is approximately $0.12–$0.15 per GB.

3. Institutional Enterprise Deployment (₹50 Lakhs+ / year)

For a "RayCtified" model to be used by a bank for compliance, the costs move from infrastructure to personnel and certification:

- Engineering Salaries: A Senior ML Engineer or MLOps specialist in India (necessary for institutional maintenance) commands a salary of ₹25 Lakhs to ₹50 Lakhs per year as of 2026.
- Platform Licensing: Professional AI model auditing solutions typically start at $50,000 (~₹42 Lakhs) annually for enterprise contracts.
- Compliance/Security: SOC2 Type II or ISO 27001 certifications and regular penetration testing can add ₹5–₹10 Lakhs in initial one-time costs.

The "Zero-Retention" Cost Advantage

Our project specifically highlights a "Zero-Retention" policy in the audit reports. This is a major cost-saving architectural choice:

- Lower Storage Costs: By not storing massive audit logs or sensitive applicant data, we avoid the high costs associated with encrypted long-term Cloud Storage.
- Reduced Liability: It significantly lowers the "cost of risk" related to data breaches or GDPR/DPDP compliance fines, which can often be the most expensive part of any institutional AI project.

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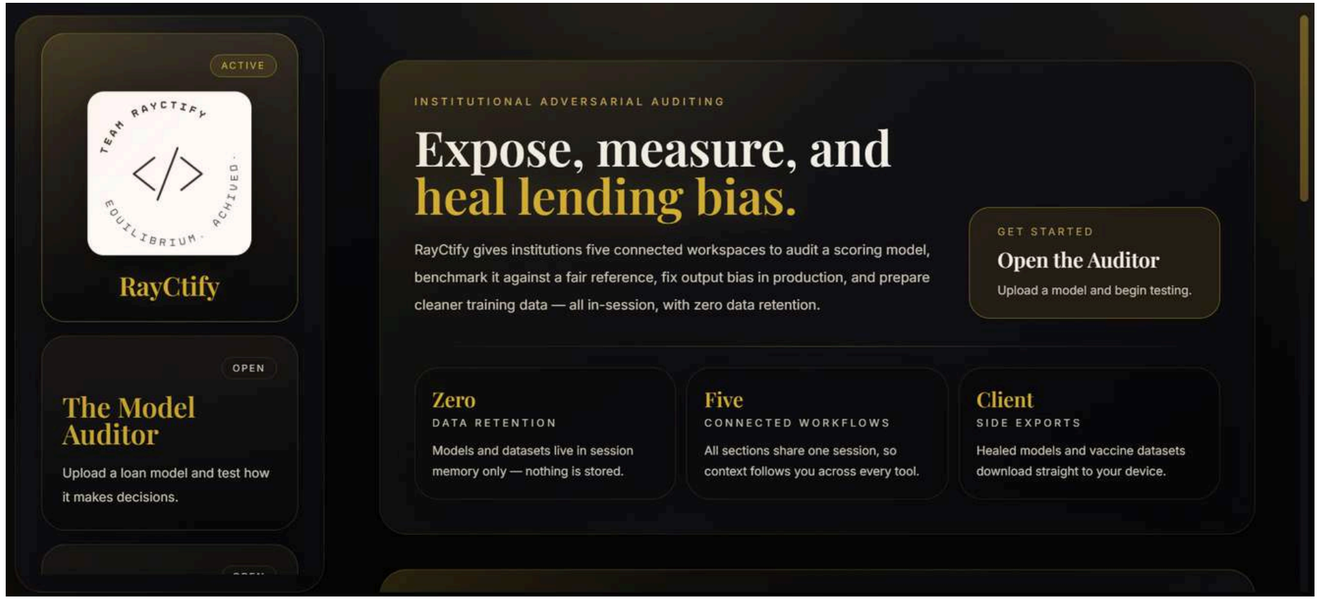
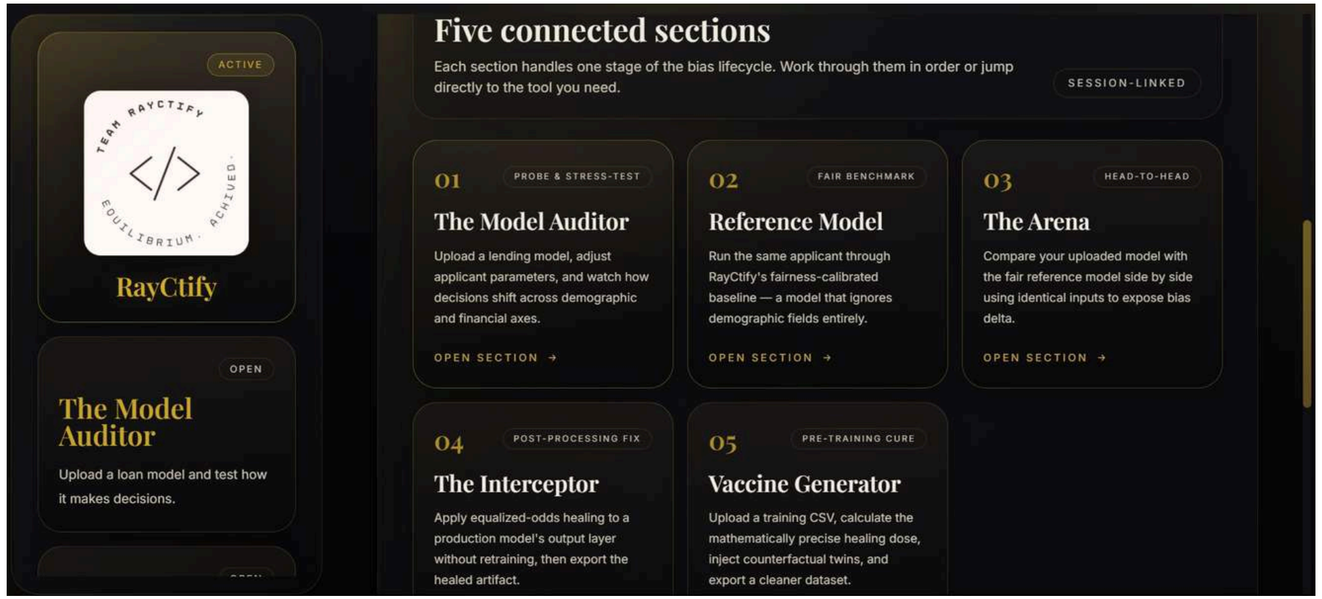
SNAP-SHOTS OF THE MVP





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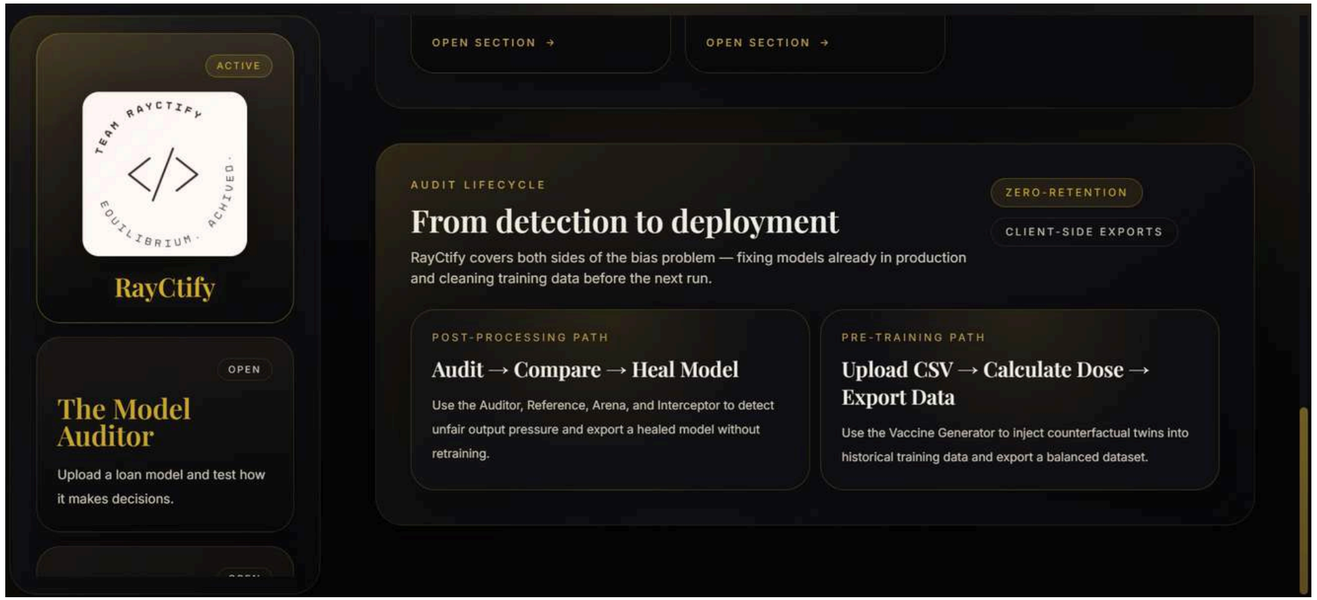

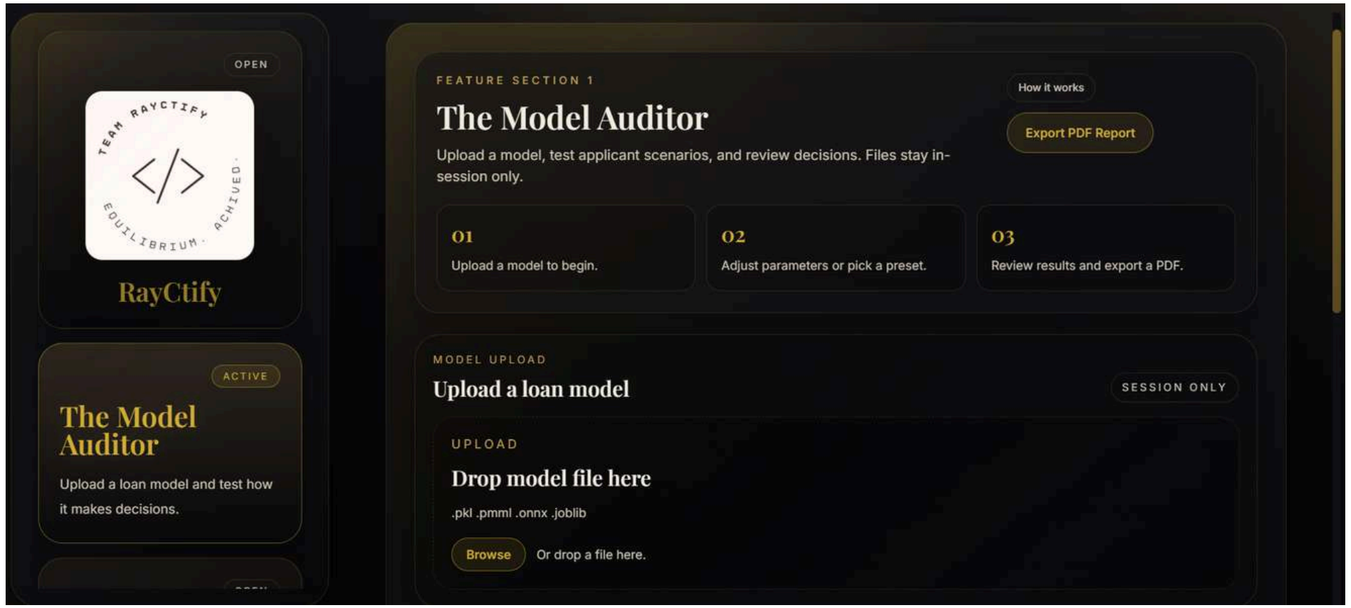

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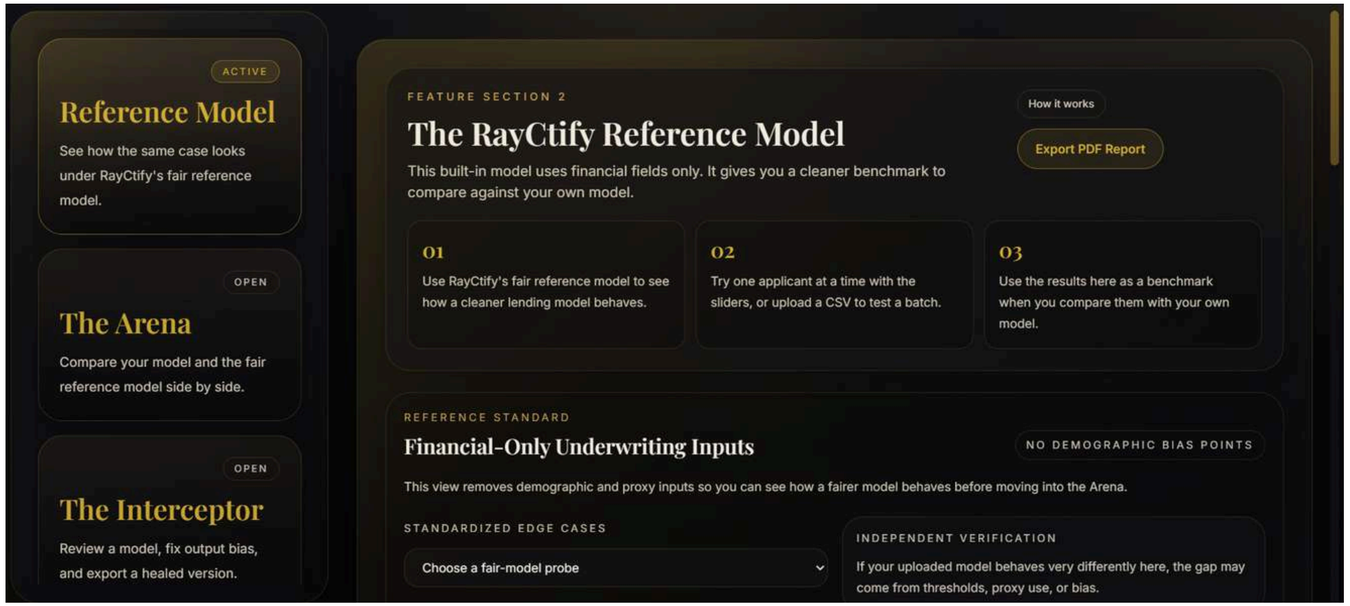

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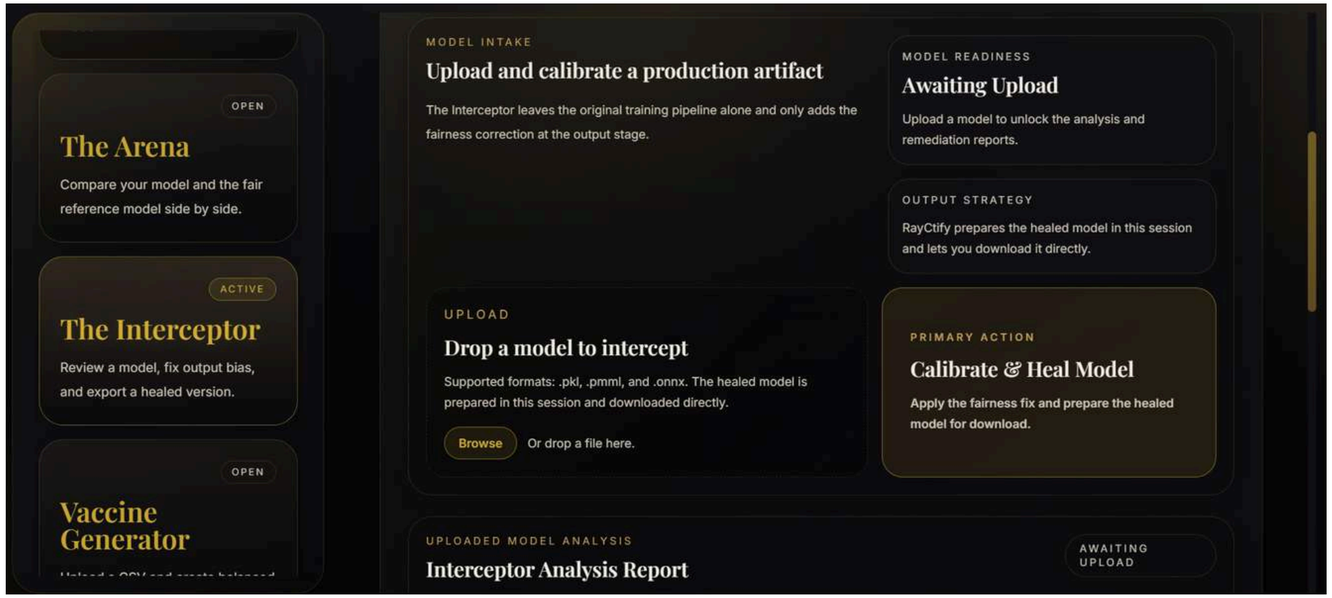

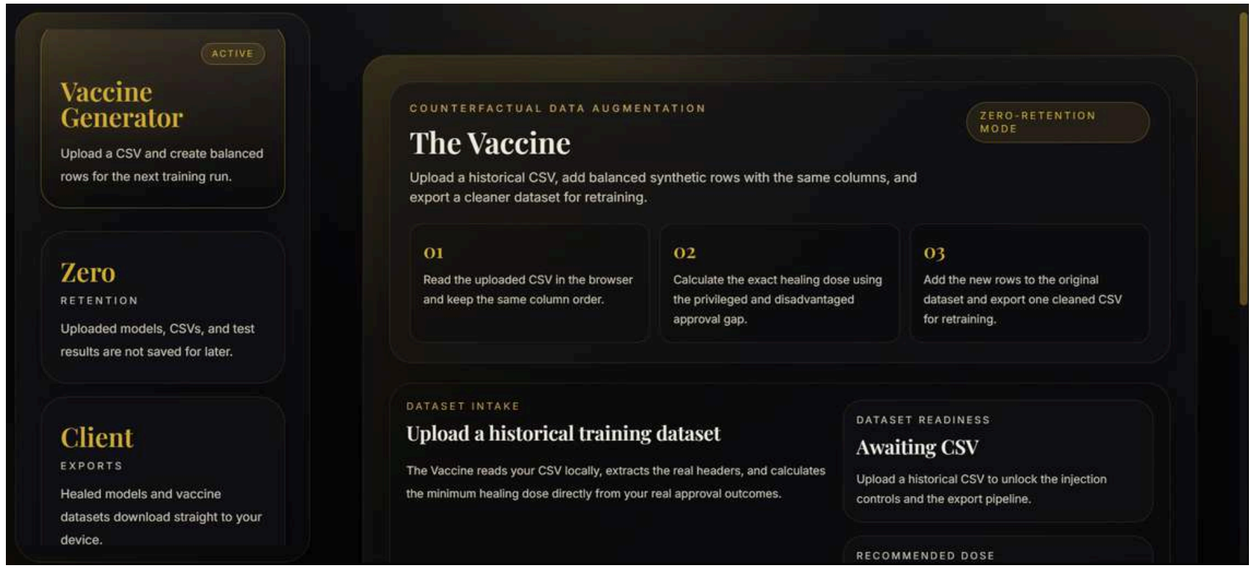

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ADDITIONAL DETAILS/FUTURE DEVELOPMENT

Advanced Fairness Techniques (In-Processing)

Currently, the system uses "Post-Mitigation" via the wrapper. The next evolution is to integrate mitigation during the training phase.

Proxy Feature Detection & Redlining Analysis

Bias often hides in non-sensitive fields. Even if "Race" is removed, a "Zip Code" or "College Name" can act as a high-correlation proxy for race or socioeconomic status.

Intersectional Fairness Auditing

Developing a specialized audit mode that checks for "Double Jeopardy"- ensuring the model is fair to specific sub-groups, such as "Black Women" or "Elderly Minorities," rather than just checking broad categories.

Explainable AI (XAI) & Narrative Generation

Moving from raw percentages to human-readable legal justifications.

Enterprise Integration (API & Monitoring)

Transforming RayCtify from a standalone auditor into a continuous monitoring suite.

LINKS TO OUR:

1. **GitHub Public Repository:** https://github.com/nahArnav/RayCtify.git

2. **Demo Video Link(s):**
   a. **Short demo video** (3 mins 44 secs - bound by the time limit given): https://drive.google.com/file/d/1E9kSmI_RZxDWE5IRgFsEGgMeaMacU1tr/view?usp=drive_link
   b. **Longer demo video** (a more detailed explanation to our solution - incase the short demo video seems insufficient and our project piques your interest :P): https://drive.google.com/file/d/1vDVfKiGt6QrfpGmWtD9N1nhXgJQU3EyJ/view?usp=drive_link

3. **MVP Link:** https://rayctify.arnavpatidar.com/

4. **Working Prototype Link:** https://rayctify.arnavpatidar.com/

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Thank You